Week 5 project pitch and (follow-up) management plan

MATH3888 (Interdisciplinary maths project unit)

Instructions for students: This document serves as your formal project pitch and a working template to help you identify and build your project as a group. It locks in your biological scope, your mathematical hypotheses, and your division of labour. Since you are working in an 8(9)-person team, successful project management is just as important as the mathematical analysis itself.

Submission: *Specific instructions for how and where to submit this finalised pitch will be distributed by Week 4.*

Note: The primary focus of this pitch is the interdisciplinary component (biology/physiology) and defining the master equations!

Part 1: Group details

- **Group number:** [Insert group #]
- **Topic domain (choose one):** Neural excitability (ch. 5), calcium dynamics (ch. 7), or neuroendocrine cells (ch. 9).
- **Working title:** [Insert a specific title for your project]

Part 2: The "buddy system" roster (4x2 matrix)

List the names of your 8(9) team members, divided into your 4 functional pairs.

Role flexibility: You have the autonomy to define the exact roles of your 4 buddy pairs based on your group's strengths. You may use the suggested default roles below (physiology, maths, computation, synthesis) or rename and redistribute the responsibilities, provided all biological research, mathematical translation, coding, and project management requirements are met.

Component	Pair responsibilities (customise as needed)	Student 1	Student 2
Pair 1: Physiology	Sourced experimental data, defined pathways, explained agonist protocols.	[Name]	[Name]
Pair 2: Maths translation	Defined master ODE equations and mapped flux terms to cell biology.	[Name]	[Name]
Pair 3: Computation	Adapting Python code for numerical experiments and setting up `PyDSTool`.	[Name]	[Name]
Pair 4: PM and synthesis	Timeline tracking, weekly meeting minutes, formatting, and synthesis.	[Name]	[Name]

(Note: If your group drops below 8 students during the semester, contact your instructor immediately to trigger a contingency structure).

Part 3: Interdisciplinary scope and hypotheses

1. The cell or signalling pathway under study

- **Physiological features:** Describe the main physiological features of your cell under study.
- **Visual requirement:** You *must* include a diagram/cartoon of your specific cell model that indicates all relevant components (e.g., specific ion channels, receptors, pumps, transporters) that you intend to model.

2. Experimental data (agonists and drugs/blockers)

- **Interventions:** Introduce all relevant agonists and drugs/blockers used in your experimental data. Provide general background about what is known regarding their effects or activity.
- **Experimental protocol:** Explicitly describe the experimental protocol that led to the data shown *without* any drugs/blockers (i.e., only with the agonist or stimulus). Then, describe the protocol that led to the data *with* the drug/blocker applied.
- **Data presentation:** Present the experimental data/findings (include the relevant tables, graphs, or time-series figures from your sourced literature).

3. Mathematical modelling and hypothesis

- **Master equations:** Write down a system of deterministic ODEs (ideally 3–5 dimensions) representing your base model. *If you use a model from the literature, cite appropriately!*
- **Flux terms:** Give a short description of all your variables and your flux/ionic current terms on the right-hand sides in relation to your specific cell model.
- **Hypothesis:** Translate your physiological hypothesis about the experimental cell signalling into a corresponding maths model hypothesis. What dynamical changes do you expect to see?

Part 4: Block 2 action plan (weeks 6–9)

To achieve a distinction/high distinction, your project must go beyond replicating textbook models. Outline what each pair will do between now and the **week 8 detailed update** to hit both the base and advanced requirements.

- **The base requirement:** Implement your system in Python and show that it is able to create the required baseline dynamics (e.g., action potentials, calcium oscillations, or bursting dynamics). Provide time traces of all variables and corresponding parameter values.
- **The advanced requirement:** Perform a bifurcation analysis (using `PyDSTool`) with respect to your agonist concentration or applied stimulus. Provide a bifurcation diagram and describe all important/relevant bifurcations observed.

Immediate next steps:

- **Pair 1 will start by:** [e.g., reviewing literature to predict exact biological tipping points when the blocker is applied.]
- **Pair 2 will start by:** [e.g., identifying which mathematical parameter in our ODE system corresponds to the drug blocker.]
- **Pair 3 will start by:** [e.g., coding the base requirement time-traces in Python to prove the model oscillates.]
- **Pair 4 will start by:** [e.g., setting up the weekly minutes template and drafting the biological introduction.]

Part 5: Internal hand-offs and risk management

Internal interfaces

Define what exactly one pair is delivering to the next to ensure the final report is ready by week 12.

Delivering	Receiving	What exactly is being handed off?	Deadline
Pair 2	**Pair 3**	*Example: finalised 3-5D master ODE equations and the chosen bifurcation parameter.*	Week 7
Pair 3	**Pair 4**	*Example: baseline Python time traces and finalised `PyDSTool` bifurcation diagrams.*	Week 10

Mandatory weekly meeting minutes

To avoid silent failures, your group must meet weekly and document your progress. **These minutes must define specific weekly goals and identify exactly who is responsible for what.** Pair 4 (or your designated PM team) is responsible for tracking these tasks. Consistent failure to deliver weekly goals will trigger administrative intervention.

Risk management contingency plan

What is your team's plan if a major bottleneck occurs (e.g., pair 3 cannot get `PyDSTool` to compile the novel equations, or the chosen paper's equations are too stiff to solve)?
Contingency: [Write 2–3 sentences explaining how your group will pivot. For example, will you revert to replicating the baseline textbook model for a pass/credit while troubleshooting the novel extension?]

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Appendix: Suggested textbook benchmark models

If your group is unsure where to begin your literature search, or if you are aiming for a solid pass/credit by analysing a baseline model, consider the following examples extracted directly from the course textbook. We have selected one benchmark model per topic domain.

All textbook references refer to: *Mathematical Physiology* by J. Keener & J. Sneyd (3rd edition).

Chapter 5: Neural excitability

- **Benchmark model: The Hodgkin-Huxley model**
  - **Source:** Section 5.1.3 (Summary of the equations, p. 241).
  - **Dimensionality:** 4D ODE (v, m, n, h).
  - **Biological focus:** The foundational model of action potentials and neural excitability.
  - **Base requirements:** Simulate current clamp experiments (I_{app} as agonist). Apply virtual tetrodotoxin (TTX) to block sodium channels, and tetraethylammonium (TEA) for potassium. Perform a 1D bifurcation diagram against I_{app} to find the Hopf bifurcation.
  - **Extensions:** Implement a two-parameter bifurcation analysis in `PyDSTool` (e.g., applied current vs. maximum channel conductance) to mathematically map the exact boundaries of the oscillatory regime, or expand the model to 5D by incorporating the binding kinetics of a novel neurotoxin or anaesthetic.

Chapter 7: Calcium dynamics

- **Benchmark model: The modal IP_3 receptor calcium model**
  - **Source:** Section 7.5.1 (A simple example, p. 358).
  - **Dimensionality:** 3D ODE (c, c_e, h).
  - **Biological focus:** Calcium oscillations in non-excitable cells (e.g., hepatocytes).
  - **Base requirements:** Use IP_3 concentration (p) as the agonist to trigger calcium spiking. Simulate thapsigargin (blocks the SERCA pump). Map the bifurcations marking the onset and cessation of oscillations.
  - **Extensions:** Compute a two-parameter bifurcation diagram (e.g., IP_3 concentration vs. total cell calcium) using numerical continuation software to reveal the global boundaries of the bursting regime and distinguish regions of complex bursting from simple spiking.

Chapter 9: Neuroendocrine bursting

- **Benchmark model: The reduced Chay-Keizer pancreatic β-cell model**
  - **Source:** Section 9.1.1.1 (Fast-slow timescale analysis: Plateau bursting, p. 473).
  - **Dimensionality:** 3D ODE (V, n, c).
  - **Biological focus:** Insulin-secreting cells that couple glucose metabolism to electrical excitability.
  - **Base requirements:** Simulate glucose as the agonist by altering the rate of calcium removal, k_c. Simulate tolbutamide (forces K_{ATP} closed) or nifedipine (blocks Ca^{2+} channels). Generate a 1D bifurcation diagram against the k_c parameter.
  - **Extensions:** Use numerical continuation software to trace bifurcations in a two-parameter plane (e.g., treating both slow calcium and the agonist as parameters) to fully map the bursting boundaries.